Exercise 11

On vector product and scalar triple product

From the school book | Understand | Apply | Higher Order Thinking Skills

1 If $\|\vec{A}\| = 8$, $\|\vec{B}\| = 5$, the measure of the angle between the two vectors $\vec{A}$ and $\vec{B}$ is 120° **Find :** $\|\vec{A} \times \vec{B}\|$ « $20\sqrt{3}$ »

2 If $\|\vec{A}\| = 5$, $\|\vec{B}\| = 8.5$, $\theta = 30°$, $\vec{C}$ is a unit vector perpendicular to each of $\vec{A}$ and $\vec{B}$

Find : $\vec{A} \times \vec{B}$ « $\pm 21.25\,\vec{C}$ »

3 If $\vec{A} \times \vec{B} = -65\,\vec{C}$ and $\|\vec{A}\| = 5$, $\|\vec{B}\| = 26$, find the measure of the angle between $\vec{A}$ and $\vec{B}$ « 30° or 150° »

4 **Find $\vec{A} \times \vec{B}$ in each of the following :**

(1) $\vec{A} = (-2, 3, 1)$, $\vec{B} = (1, 3, -4)$ (2) $\vec{A} = -\hat{i} - 2\hat{j}$, $\vec{B} = 3\hat{j} - 5\hat{k}$

5 If $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = 3\hat{i} - 2\hat{j} + 5\hat{k}$

Find : (1) $\vec{A} \times \vec{B}$ (2) $(2\vec{A}) \times (3\vec{B})$ (3) $\vec{A} \times (\vec{A} - 2\vec{B})$

6 If $\vec{A} = 4\hat{i} + 3\hat{j} + \hat{k}$, $\vec{B} = \hat{i} - 2\hat{k}$, find : $\|2\vec{B} \times \vec{A}\|$ « $6\sqrt{14}$ »

7 Find sine of the included angle between the two vectors $\vec{A} = 3\hat{i} - 4\hat{j} + 5\hat{k}$, $\vec{B} = \hat{i} - \hat{j} + \hat{k}$ « $\frac{1}{5}$ »

8 **If $\vec{A} = (2, 3, 0)$, $\vec{B} = (-1, 3, 1)$, $\vec{C} = (1, 2, 5)$ Find :**

(1) $\vec{A} \times \vec{B}$ (2) $\vec{C} \times \vec{C}$ (3) $(\vec{A} - \vec{B}) \times (\vec{C} - \vec{A})$

9 **Choose the correct answer from those given :**

(1) If $\vec{A} = (3, 0, 4)$, $\vec{B} = \hat{i} - 2\hat{j} + 3\hat{k}$, then $\vec{A} \times \vec{B}$ = ...

(a) $(8, -5, -6)$ (b) $(8, 5, 6)$ (c) $(8, -5, 6)$ (d) $(-8, -5, -6)$

(2) If $\vec{A} = \hat{i} - 3\hat{j} + \hat{k}$, $\vec{B} = \hat{i} + \hat{j} + \hat{k}$, then $\|\vec{A} \times \vec{B}\|$ = ...

(a) $4\sqrt{2}$ (b) $3\sqrt{2}$ (c) $2\sqrt{5}$ (d) $2\sqrt{3}$

(3) If $\vec{A}$, $\vec{B}$ are non-zero vectors and the measure of the angle between them is 30° , $\|\vec{A}\| = 5$, $\|\vec{B}\| = 4$, then $\|\vec{A} \times \vec{B}\|$ = ...

(a) $20\,\vec{C}$ (b) $10\,\vec{C}$ (c) 10 (d) 20

(4) If $\vec{A} \parallel \vec{B}$, then $\vec{A} \times \vec{B}$ = ...

(a) $\vec{O}$ (b) 1 (c) $\|\vec{A}\|$ (d) $\|\vec{B}\|$

(5) If $\|\vec{A} \times \vec{B}\| = 36\sqrt{3}$, $\|\vec{A}\| = 8$, $\|\vec{B}\| = 9$, then the measure of the angle between $\vec{A}$ and $\vec{B}$ could be .....

(a) 30° (b) 120° (c) 150° (d) 45°

(6) If $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} - \hat{k}$, then $\vec{A} \times (\vec{A} - \vec{B})$ = ...

(a) $\hat{i} + \hat{k}$ (b) $-3\hat{j} + 3\hat{k}$ (c) $-3\hat{i} - 3\hat{j}$ (d) $3\hat{i} - 2\hat{j}$

(7) If the vectors $(2 , k , -3)$, $(4 , 6 , -6)$ are parallel , then k = ...

(a) 6 (b) 3 (c) – 3 (d) 1

(8) If $\vec{A} = (4 , -k , 6)$, $\vec{B} = (2 , 2 , m)$ and $\vec{A} \parallel \vec{B}$, then k + m = ...

(a) – 3 (b) – 2 (c) – 1 (d) zero

(9) $\hat{i} \times \hat{j}$ = ...

(a) $\vec{0}$ (b) zero (c) 1 (d) $\hat{k}$

(10) If $\{\hat{i} , \hat{j} , \hat{k}\}$ is a set of right-hand system of unit vectors , then .....

(a) $\hat{i} . \hat{j} = 1$ (b) $\hat{i} . \hat{i} = 1$ (c) $\hat{i} \times \hat{j} = 1$ (d) $\hat{i} \times (\hat{j} \times \hat{k}) = 1$

(11) If $\vec{A} = (4 , -5 , 1)$, $\vec{B} = (2 , -k , -2)$, $\vec{C} = (-4 , 4 , m - 2)$ and $\overrightarrow{AB} \parallel \vec{C}$, then k – m = ...

(a) 5 (b) 7 (c) 8 (d) 9

(12) If $\vec{A} = (1 , 0 , 2)$, $\vec{B} = (2 , -1 , -2)$, then $(\vec{A} \times \vec{B}) . (\vec{B} \times \vec{A})$ = ...

(a) 41 (b) 31 (c) – 31 (d) – 41

(13) If $\vec{A}$, $\vec{B}$ are two unit perpendicular vectors , then $\|(\vec{A} - \vec{B}) \times (\vec{A} + \vec{B})\|$ = ...

(a) zero (b) 1 (c) – 1 (d) 2

(14) If $\|\vec{A} \times \vec{B}\| = 4$, then the unit vector perpendicular to each of $\vec{A}$ and $\vec{B}$ = ...

(a) $\pm \frac{1}{8} (\vec{A} \times \vec{B})$ (b) $\pm \frac{1}{4} (\vec{A} \times \vec{B})$ (c) $\pm \frac{1}{2} (\vec{A} \times \vec{B})$ (d) $\pm 4 (\vec{A} \times \vec{B})$

(15) ***(Trial 2021)*** If $\{\hat{i} , \hat{j} , \hat{k}\}$ is a set of a right hand system of unit vectors , then $(\hat{i} \times \hat{j}) . \hat{k} + \hat{i} . \hat{j}$ = ...

(a) – 1 (b) zero (c) 1 (d) 2

(16) If $\vec{A} \times \vec{B} = \vec{C}$, then .....

(a) $\vec{B} \times \vec{C} = \vec{A}$ (b) $\vec{A} \times \vec{C} = \vec{B}$

(c) $\vec{A} \times (\vec{B} + \vec{A}) = \vec{C}$ (d) $\vec{B} \times \vec{A} = \vec{C}$

(17) If $\vec{A}$ is a non-zero vector , then the expression

$\| \vec{A} \times \hat{i} \|^2 + \| \vec{A} \times \hat{j} \|^2 + \| \vec{A} \times \hat{k} \|^2 =$ .....

(a) $\| \vec{A} \|^2$ (b) zero (c) $2 \| \vec{A} \|^2$ (d) $3 \| \vec{A} \|^2$

(18) If $\vec{A}$, $\vec{B}$ are two non-zero parallel vectors , then which of the following statements is not true ?

(a) $\vec{A} = k \vec{B}$, $k \neq 0$ (b) $\frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z}$ (c) $\vec{A} \cdot \vec{B} =$ zero (d) $\vec{A} \times \vec{B} = \overrightarrow{zero}$

(19) If $\vec{C}$ is a unit vector perpendicular to $\vec{A}$ and $\vec{B}$, $\| \vec{A} \| = 9$, $\| \vec{B} \| = 16$ and $\vec{A} \cdot \vec{B} = -72\sqrt{3}$, then $\vec{A} \times \vec{B} =$ .....

(a) $\pm 26 \vec{C}$ (b) $\pm 36 \vec{C}$ (c) $\pm 72 \vec{C}$ (d) $\pm 72\sqrt{3} \vec{C}$

(20) *(1st Session 2021)* If $\| \vec{A} \| = \sqrt{13}$, $\vec{A} \,//\, \overrightarrow{DC}$ and in the same direction , such that D (1 , 3 , – 2) , C (1 , – 1 , 4) and $\vec{B} = (-2 , 3 , 5)$, then $\vec{A} \times \vec{B} =$ .....

(a) $-19 \hat{i} + 6 \hat{j} + 4 \hat{k}$ (b) $-28 \hat{i} - 12 \hat{j} - 8 \hat{k}$

(c) $-28 \hat{i} + 12 \hat{j} - 8 \hat{k}$ (d) $-19 \hat{i} - 6 \hat{j} - 4 \hat{k}$

(21) *(1st Session 2022)* If $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{E}$ are four vectors in the same plane , and $\vec{C} = \vec{A} + \vec{B}$, $\vec{E} = \vec{A} - \vec{B}$, if $\vec{C} \times \vec{E} = 3 \vec{e}$, where $\vec{e}$ is a perpendicular unit vector to the plane containing the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{E}$, then $\vec{A} \times \vec{B} =$ .....

(a) $-\frac{3}{2} \vec{e}$ (b) $\frac{3}{2} \vec{e}$ (c) $\frac{2}{3} \vec{e}$ (d) $-\frac{2}{3} \vec{e}$

(22) *(2nd Session 2022)* If $\{\hat{i} , \hat{j} , \hat{k}\}$ is the set of right hand system of unit vectors and $\vec{A} = \hat{i} \times 2 \hat{j} + 2 \hat{i} \times \hat{k} + \hat{j} \times \hat{k}$, then the unit vector in the direction of $\vec{A}$ is .....

(a) $\left(\frac{-2}{3} , \frac{1}{3} , \frac{2}{3}\right)$ (b) $\left(\frac{1}{3} , \frac{2}{3} , \frac{-2}{3}\right)$ (c) $\left(\frac{2}{3} , \frac{-2}{3} , \frac{1}{3}\right)$ (d) $\left(\frac{1}{3} , \frac{-2}{3} , \frac{2}{3}\right)$

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10 **Find the result :** $[(1 , -2 , 3) \times (2 , 1 , -3)] \times (-3 , 1 , 2)$

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11 If $\| \vec{A} \| = 6$ and the direction cosines of the vector $\vec{A}$ are $\frac{2}{3} , \frac{-2}{3} , \frac{1}{3}$ respectively and $\vec{B} = (-2 , 3 , 5)$, find : $\vec{A} \times \vec{B}$

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12 ABCD is a square of side length 12 cm. , $\vec{C}$ is a unit vector perpendicular to its plane , **find :**

(1) $\overrightarrow{AB} \times \overrightarrow{CA}$ **(2)** $\overrightarrow{BD} \times \overrightarrow{AC}$ **(3)** $\overrightarrow{AB} \times \overrightarrow{BC}$

13 ABCD is a rectangle , in which AB = 3 cm. , BC = 4 cm. its diagonals intersect at M , $\vec{C}$ is a unit vector perpendicular to its plane , **find :**

(1) $\overrightarrow{AB} \times \overrightarrow{AC}$ | (2) $\overrightarrow{AD} \times \overrightarrow{BC}$ | (3) $\overrightarrow{AB} \times \overrightarrow{BD}$

(4) $\overrightarrow{CM} \times \overrightarrow{CB}$ | (5) $\overrightarrow{AB} \times \overrightarrow{CB}$ | (6) $\overrightarrow{DM} \times \overrightarrow{DC}$

14 **Find a unit vector perpendicular to each $\vec{A}$ and $\vec{B}$ where :**

(1) $\vec{A} = 4\hat{i} - 6\hat{j} + 2\hat{k}$, $\vec{B} = -2\hat{i} + 4\hat{j} - 6\hat{k}$

(2) $\vec{A} = \hat{i} - 2\hat{j} + 2\hat{k}$, $\vec{B} = 3\hat{i} - 4\hat{j}$

15 If A (0 , 0 , 1) , B (1 , 0 , 0) , C (0 , 1 , 0) , find the orthogonal unit vector to the plane ABC

16 Find the vector $\vec{D}$ that is perpendicular to the plane which contains the three points A (1 , – 1 , 4) , B (2 , 0 , 1) , C (0 , 2 , 3)

17 Find the vector whose magnitude 3 units and perpendicular to each of the two vectors $\vec{A} = 3\hat{i} + \hat{j} - 4\hat{k}$, $\vec{B} = 6\hat{i} + 5\hat{j} - 2\hat{k}$

18 Prove that the vector $\vec{A} \times \vec{B}$ is perpendicular to each of the two vector $\vec{A}$, $\vec{B}$ where $\vec{A}$ = (1 , 3 , – 1) , $\vec{B}$ = (2 , 0 , 1) using the scalar triple product.

19 Find : L , m if : $(2\hat{i} + 6\hat{j} - 27\hat{k}) \times (\hat{i} + L\hat{j} + m\hat{k}) = \vec{O}$ « 3 , $\frac{-27}{2}$ »

20 **Calculate the area of the parallelogram LMNE in each of the following :**

(1) L (1 , 1) , M (2 , 3) , N (5 , 4)

(2) L (2 , 1 , 3) , M (1 , 4 , 5) , N (2 , 5 , 3) « 5 , $4\sqrt{5}$, area unit »

21 **Calculate the area of the triangle ABC in each of the following :**

(1) A (5 , 1 , – 2) , B (4 , – 4 , 3) , C (2 , 4 , 0)

(2) A (4 , 0 , 2) , B (2 , 1 , 5) , C (– 1 , 0 , – 1) « 16.7 , $\frac{5}{2}\sqrt{19}$, area unit »

22 If $\vec{A}$ = (– 3 , 1 , 2) , $\vec{B}$ = (3 , 4 , – 1) find the area of parallelogram in which $\vec{A}$, $\vec{B}$ are two adjacent sides. « $3\sqrt{35}$ area unit »

23 Find the area of triangle in which $\vec{A}$, $\vec{B}$ are two adjacent sides where $\vec{A} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = \hat{i} - 3\hat{j} + 4\hat{k}$ « $5\sqrt{3}$ area unit »

24 📖 **In each of the following , show whether the two given vectors are parallel or perpendicular or otherwise :**

(1) $\vec{A} = (0 , 2 , 3)$, $\vec{B} = (3 , 0 , -4)$

(2) $\vec{A} = -2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = 8\hat{i} - 4\hat{j} + 8\hat{k}$

25 **Prove that the points :**

A (5 , 6 , 7) , B (7 , – 8 , 9) , C (3 , 20 , 5) are on the same straight line.

26 If $\vec{A} = (1 , 6 , 2)$, $\vec{B} = (k , 3 , m)$, $\vec{C} = (k , m , k + m)$ and $\vec{A} // \vec{B}$, find : $\| \vec{C} \|$ « $\frac{\sqrt{14}}{2}$ »

27 **If $\vec{A} = (2 , 3 , 0)$, $\vec{B} = (m , 2 , 0)$, determine the value of m in the following cases :**

(1) $\vec{A} \times \vec{B} = -5\hat{k}$ (2) $\vec{A} \perp \vec{B}$

(3) The measure of the angle between $\vec{A}$ and $\vec{B} = 45°$ « 3 , – 3 , $\frac{-2}{5}$ or 10 »

28 📖 If $\vec{A} = (2 , -3)$ and $\vec{B} // \vec{A}$ and if $\| \vec{B} \| = 3\sqrt{13}$ Find : $\vec{B}$ « $\pm (6 , -9)$ »

29 *(2nd Session 2018)* ABCD is a quadrilateral in which A (3 , 0 , 2) , B (6 , 2 , 5) , C (4 , 4 , 5) and D (1 , 2 , 2)

(1) Prove that the figure ABCD is a parallelogram and find its area.

(2) Find the unit vector perpendicular to the plane of the quadrilateral.

« $2\sqrt{43}$ square unit , $\pm \left(\frac{-3}{\sqrt{43}} , \frac{-3}{\sqrt{43}} , \frac{5}{\sqrt{43}} \right)$ »

30 **Choose the correct answer from those given :**

(1) If θ is the angle between the two vectors $(\hat{i} + \hat{j} + \hat{k})$ and $(2\hat{i} - \hat{j} + 2\hat{k})$, then sin θ = ...

(a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{\frac{2}{3}}$

(c) $\frac{\sqrt{2}}{3}$ (d) non of the previous.

(2) 📖 If $\vec{A} \times \vec{B} = \vec{O}$, $\vec{A} . \vec{C} = 0$, then $\vec{B} . \vec{C}$ = ...

(a) 0 (b) 1 (c) $\vec{A}$ (d) $\| \vec{B} \|$

(3) The unit vector perpendicular to each of vectors $(\hat{i} + \hat{j})$ and $(\hat{j} + \hat{k})$ is .....

(a) $\hat{i} - \hat{j} + \hat{k}$ (b) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$ (c) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ (d) $\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

(4) If $\vec{A} = 3\hat{i} - 6\hat{j} - \hat{k}$, $\vec{B} = \hat{i} + 4\hat{j} - 3\hat{k}$ and $\vec{C} = 3\hat{i} - 4\hat{j} - 12\hat{k}$, then the projection of the vector $(\vec{A} \times \vec{B})$ on the vector $\vec{C}$ is .....

(a) 14 (b) – 14 (c) 12 (d) 15

(5) If $\vec{A} \cdot \vec{B} = \| \vec{A} \times \vec{B} \|$, then the measure of the angle between the two vectors $\vec{A}$ and $\vec{B}$ equals .....

(a) 30° (b) 45° (c) 60° (d) 135°

(6) The area of Δ ABC is 24 cm^2 , then $\| \overrightarrow{BA} \times \overrightarrow{BC} \|$ = ...

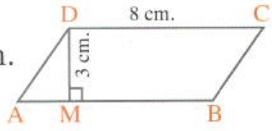
(a) 12 (b) 24 (c) 36 (d) 48

(7) If the area of the parallelogram ABCD equals 12 cm^2 , then $\| \overrightarrow{AC} \times \overrightarrow{BD} \|$ = ...

(a) 6 (b) 12 (c) 24 (d) 48

(8) ***(2nd Session 2024)*** **In the opposite figure :**

ABCD is a parallelogram in which , $\overline{DM} \perp \overline{AB}$, if DM = 3 cm. , DC = 8 cm. , then $\| \overrightarrow{AB} \times \overrightarrow{AD} \|$ = ... cm^2



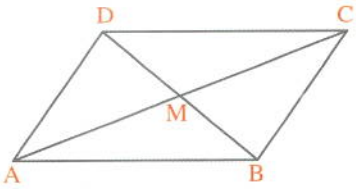
(a) 48 (b) 24 (c) 12 (d) 6

(9) If ABCD is a parallelogram , M is the point of intersection of its diagonals , then $\| \overrightarrow{MD} \times \overrightarrow{MC} \|$ = ...

(a) $\vec{O}$ (b) $\| \overrightarrow{AB} \times \overrightarrow{AD} \|$ (c) $\| \overrightarrow{MC} \times \overrightarrow{MB} \|$ (d) $\overrightarrow{MD} \cdot \overrightarrow{MC}$

(10) ***(1st Session 2024)*** **In the opposite figure :**

ABCD is a parallelogram its diagonals intersect at M , if $\| \overrightarrow{AC} \times \overrightarrow{BD} \|$ = 24 square unit , then $\| \overrightarrow{MB} \times \overrightarrow{MC} \|$ = ... square unit.



(a) 3 (b) 4 (c) 6 (d) 12

(11) ***(2nd Session 2019)*** ABCD is a parallelogram in which $\overrightarrow{AB}$ = (2 , 2 , – 1) , $\overrightarrow{AD}$ = (– 1 , 2 , – 3) , then the surface area of the parallelogram = ... square units.

(a) 6 (b) $7\sqrt{2}$ (c) $3\sqrt{11}$ (d) $\sqrt{101}$

(12) Area of parallelogram in which its diagonals are $\vec{M} = 2\hat{i} - \hat{j}$ and $\vec{N} = 4\hat{i} - 5\hat{j}$ equals ..... square unit.

(a) $3\sqrt{2}$ (b) $\frac{3\sqrt{2}}{2}$ (c) $6\sqrt{2}$ (d) 3

(13) Area of parallelogram ABCD whose diagonals intersect at M equals .....

(a) $\| \overrightarrow{AM} \times \overrightarrow{BM} \|$ (b) $\| \overrightarrow{AB} \times \overrightarrow{AM} \|$ (c) $\| \overrightarrow{AB} \times \overrightarrow{AC} \|$ (d) $\| \overrightarrow{AC} \times \overrightarrow{BD} \|$

(14) If $\overrightarrow{AD}$ is a median of Δ ABC , then $\| \overrightarrow{AD} \times \overrightarrow{AB} \|$ = ... × the area of Δ ABC

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 2

(15) If $\vec{A} = (1 , -1 , 2)$, $\vec{B} = (3 , -2 , 0)$, $\vec{C} = (0 , 2 , 4)$, then $\vec{A} . \vec{B} \times \vec{C}$ = ...

(a) 10 (b) 12 (c) 14 (d) 16

(16) $\vec{A} . (\vec{B} \times \vec{C})$ = ...

(a) $\vec{A} . (\vec{C} \times \vec{B})$ (b) $\vec{C} . (\vec{B} \times \vec{A})$ (c) $\vec{A} \times \vec{B} \times \vec{C}$ (d) $\vec{C} . (\vec{A} \times \vec{B})$

(17) $\vec{B} . (\vec{A} \times \vec{C})$ = ...

(a) $(\vec{B} \times \vec{A}) . \vec{C}$ (b) $\vec{A} . (\vec{B} \times \vec{C})$ (c) $(\vec{A} \times \vec{B}) \times \vec{C}$ (d) $\vec{C} . (\vec{A} \times \vec{C})$

(18) $\hat{i} . (\hat{j} \times \hat{k}) + \hat{j} . (\hat{k} \times \hat{i}) + \hat{k} . (\hat{i} \times \hat{j})$ = ...

(a) 1 (b) 3 (c) – 3 (d) zero

(19) The volume of the parallelepiped in which 3 adjacent edges represented by the vectors $\vec{A} = (3 , -4 , 0)$, $\vec{B} = (0 , -4 , 3)$, $\vec{C} = (0 , 0 , 5)$ equals ..... cubic unit.

(a) 12 (b) 50 (c) 60 (d) 125

(20) *(2nd Session 2023)* If $\vec{A} = (m , -3 , 1)$, $\vec{B} = (1 , 0 , -2)$ and $\vec{C} = (-2 , -1 , 4)$ represent three adjacent edges in a parallelepiped whose volume equals 7 cubic units , then m = ...

(a) – 4 or 3 (b) – 3 or 4 (c) 2 or – 2 (d) 5 or – 1

(21) *(1st Session 2023)* If the vectors $\vec{A} = m \hat{i} + 3 \hat{j} - 2 \hat{k}$, $\vec{B} = \ell \hat{j} + 5 \hat{k}$ and $\vec{C} = n \hat{k}$ represent three adjacent edges in parallelepiped whose volume equal 40 cubic units , then m ℓ n = ± .....

(a) 10 (b) 40 (c) 20 (d) 80

(22) *(Trial 2021)* If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three connected edges in a parallelepiped where $\| \vec{A} \| = 2$ The direction cosines of the vector $\vec{A}$ are (135° , 60° , 120°) and $\vec{B} = (1 , \sqrt{2} , 0)$, $\vec{C} = (\sqrt{2} , 3 , 5)$, then the volume of the parallelepiped = ... cubic units.

(a) 16 (b) $6\sqrt{2}$ (c) 11 (d) $16\sqrt{2}$

(23) *(1st Session 2018)* In the cartesian plane $\mathcal{X}$ y if θ is the measure of the angle between $\vec{A}$ and $\vec{B}$, then $\frac{\|\vec{A} \times \vec{B}\|}{\vec{A} \cdot \vec{B}}$ = ...

(a) $\sin\theta$ (b) $\cos\theta$ (c) $\tan\theta$ (d) $\cot\theta$

(24) If $\|\vec{A} \times \vec{B}\|^2 + (\vec{A} \cdot \vec{B})^2 = 144$, $\|\vec{A}\| = 4$, then $\|\vec{B}\|$ = ...

(a) 16 (b) 8 (c) 3 (d) 5

(25) If $\vec{A}$, $\vec{B}$ are two non-zero vectores , θ is the measure of the angle between them , then $\|\vec{A} \times \vec{B}\|^2 + (\vec{A} \cdot \vec{B})^2$ = ...

(a) $a^2 b^2$ (b) 1 (c) $a^2 b^2 \sin\theta$ (d) 0

(26) If $\vec{A}$, $\vec{B}$ are two unit vectors , θ is the measure of the angle between them , then $2\|\vec{A} \times \vec{B}\|(\vec{A} \cdot \vec{B})$ = ...

(a) $\sin\theta\cos\theta$ (b) $2\sin\theta$ (c) $\cos 2\theta$ (d) $\sin 2\theta$

(27) If $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$, then .....

(a) $\vec{B} = \vec{C}$ (b) $\vec{A}$ and $\vec{B}$ are parallel.

(c) $\vec{A}$, $\vec{B}$, $\vec{C}$ are mutually perpendicular. (d) $\vec{A}$, $\vec{B}$, $\vec{C}$ in the same plane.

(28) *(1st Session 2019)* If $\vec{A} \perp \vec{B}$, $\vec{A} \perp \vec{C}$, $\vec{B} = (2, 3, 2)$, $\vec{C} = (1, 2, 1)$ and $\|\vec{A}\| = 4\sqrt{2}$, then $\vec{A}$ may be equal .....

(a) (2 , 3 , 1) (b) (– 4 , 0 , 4) (c) (4 , 4 , 0) (d) (0 , – 4 , 4)

(29) If $\vec{A} = (2, 1, -2)$, $\vec{A} + \vec{B} = \vec{A} \times \vec{B}$, then $\vec{B}$ = ...

(a) (2 , – 1 , – 2) (b) (2 , 1 , – 2) (c) (– 2 , – 1 , 2) (d) (– 2 , – 1 , 3)

(30) If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{c} = \vec{j} - \vec{k}$, $\vec{b}$ are three vectors in the space and $\vec{a} \times \vec{b} = \vec{c}$, then $\vec{b} \cdot (\vec{c} \times \vec{a})$ = ...

(a) 1 (b) 2 (c) 3 (d) 4

(31) *(Trial 2023)* ABC is a triangle whose vertices are A (4 , 0 , 0) , B (0 , m , 0) and C (0 , 0 , 2) if the area of triangle ABC = 6 square units , then m = ... where $m \in \mathbb{R}$

(a) ± 2 (b) ± 4 (c) – 2 (d) 4

(32) If $\vec{A}$, $\vec{B}$, $\vec{C}$ are unit vectors , then the greatest value for the expression $\vec{A} \cdot \vec{B} \times \vec{C}$ equals ..... cubic unit.

(a) 1 (b) 3 (c) $\sqrt{3}$ (d) $\sqrt{2}$

(33) $\vec{A} \cdot (\vec{B} + \vec{C}) \times (\vec{A} + \vec{B} + \vec{C}) =$ .....

(a) zero (b) 1 (c) $\vec{A} \cdot \vec{B} \times \vec{C}$ (d) $\vec{A} \cdot (\vec{B} + \vec{C})$

(34) If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three non-zero vectors , then $\vec{A} \cdot \vec{B} \times \vec{C} = \|\vec{A}\| \|\vec{B}\| \|\vec{C}\|$ if .....

(a) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{C} =$ zero (b) $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C} =$ zero

(c) $\vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{C} =$ zero (d) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{C} = \vec{C} \cdot \vec{A} =$ zero

(35) If $\vec{A}$, $\vec{B}$, $\vec{C}$ are unit vectors and $\vec{A} \cdot \vec{B} = \vec{A} \cdot \vec{C} = 0$, measure of the angle between $\vec{B}$ and $\vec{C}$ equals $\frac{\pi}{4}$, then $\vec{A}$ = ...

(a) $\pm 2 (\vec{B} \times \vec{C})$ (b) $\pm 3 (\vec{B} \times \vec{C})$ (c) $\pm\sqrt{2} (\vec{B} \times \vec{C})$ (d) $\pm\sqrt{3} (\vec{B} \times \vec{C})$

(36) The points A , B , C , D are coplanar , then .....

(a) $\vec{A} \cdot (\vec{B} \times \vec{C}) =$ zero (b) $\overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD}) =$ zero

(c) $\vec{A} \times \vec{B} = \vec{C} \times \vec{D}$ (d) $\overrightarrow{AB} \cdot (\overrightarrow{BC} \times \overrightarrow{BD}) \neq$ zero

(37) Volume of the triangular pyramid in which the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ represents three non-coplanar edges equals ..... cubic units.

(a) $\frac{1}{2} |\vec{A} \cdot \vec{B} \times \vec{C}|$ (b) $\frac{1}{3} |\vec{A} \cdot \vec{B} \times \vec{C}|$

(c) $\frac{1}{4} |\vec{A} \cdot \vec{B} \times \vec{C}|$ (d) $\frac{1}{6} |\vec{A} \cdot \vec{B} \times \vec{C}|$

(38) A parallelepiped formed from the vectors $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = 5\hat{i} + 2\hat{j} + \ell\hat{k}$, $\vec{C} = \ell\hat{i} + 4\hat{j} + \hat{k}$, then smaller volume of this parallelepiped at ℓ = ...

(a) 1 (b) 2 (c) 3 (d) 4

31 If $\vec{A} = (1 , 1 , 3)$, $\vec{B} = (2 , 1 , 4)$, $\vec{C} = (5 , 1 , -2)$ Find : $\vec{A} \cdot \vec{B} \times \vec{C}$ « 9 »

32 If $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{C} = \hat{i} + \hat{j} + \hat{k}$ Find : $(\vec{A} \times \vec{B}) \cdot (\vec{A} \times \vec{C})$ « – 74 »

33 If $\vec{A} = \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$, $\vec{B} = \frac{3}{7}\hat{i} - \frac{6}{7}\hat{j} + \frac{2}{7}\hat{k}$, $\vec{C} = \frac{6}{7}\hat{i} + \frac{2}{7}\hat{j} - \frac{3}{7}\hat{k}$, prove that $\vec{A}$, $\vec{B}$, $\vec{C}$ are unit vectors perpendicular to each other then , prove that : $\vec{A} \times \vec{B} = \vec{C}$

34 Find the volume of the parallelepiped in which three non parallel edges are represented by the vectors $\vec{A} = (3 , -4 , 1)$, $\vec{B} = (0 , 2 , -3)$, $\vec{C} = (3 , 2 , 2)$ « 60 volume unit »

35 *(1st Session 2018)* If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three adjacent edges in a parallelepiped such that :

$\vec{A} = (1 , 4 , 2)$, $\vec{B} (-3 , 2 , 1)$, $\vec{C} = (-1 , 1 , 4)$

Find : (1) The volume of the parallelepiped.

(2) The height of the parallelepiped drawn on the base determined by the two vectors $\vec{A}$, $\vec{B}$ « 49 cube unit , $\frac{7\sqrt{5}}{5}$ length unit »

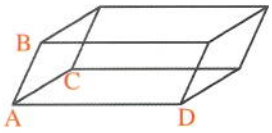
36 The volume of a parallelepiped is 546 and three adjacent edges are represented by the vectors $(-12\,\hat{i} + \ell\,\hat{k})$, $(3\,\hat{j} - \hat{k})$, $(2\,\hat{i} + \hat{j} - 15\,\hat{k})$ Find ℓ « 179 or – 3 »

37 **In the opposite figure :**

If A (1 , 1 , 1) , B (2 , 1 , 3)

, C (3 , 2 , 2) , D (3 , 3 , 4)

Find the volume of the parallelepiped. « 5 volume unit »



38 Prove that the vectors :

$\vec{A} = \hat{i} - 2\,\hat{j} + 3\,\hat{k}$, $\vec{B} = -2\,\hat{i} + 3\,\hat{j} - 4\,\hat{k}$, $\vec{C} = \hat{i} - 3\,\hat{j} + 5\,\hat{k}$ lie in the same plane.

39 Let A (4 , 8 , 12) , B (2 , 4 , 6) , C (3 , 5 , 4) , D (5 , 8 , 5)

Prove that the points A , B , C , D lie in the same plane.

40 Find the value of ℓ to make the vectors $\vec{A} = (1 , -1 , 1)$, $\vec{B} = (3 , 1 , 2)$, $\vec{C} = (1 , \ell , -3)$ are in the same plane. « 15 »

41 Find k which makes the points A , B , C , D are in the same plane where

A (3 , – 2 , – 1) , B (2 , 3 , – 4) , C (– 1 , 1 , 2) , D (4 , 5 , k) « $\frac{-146}{17}$ »

42 **Prove each of the following where $\vec{A}$, $\vec{B} \in \mathbb{R}^3$:**

(1) $\| \vec{A} \times \vec{B} \|^2 + | \vec{A} . \vec{B} |^2 = \| \vec{A} \|^2 \| \vec{B} \|^2$

(2) If $\vec{A} . \vec{B} = 0$, $\vec{A} \times \vec{B} = \vec{O}$ then $\vec{A} = \vec{O}$ or $\vec{B} = \vec{O}$

43 **Prove that :** $(\vec{A} - \vec{B}) \times (\vec{A} + \vec{B}) = 2\,(\vec{A} \times \vec{B})$

44 If $\vec{A} \times \vec{B} = \vec{C} \times \vec{D}$, $\vec{A} \times \vec{C} = \vec{B} \times \vec{D}$

Prove that the vector $(\vec{A} - \vec{D})$ and the vector $(\vec{B} - \vec{C})$ are parallel where A , B , C , D are non-zeroe vectors.

45 **Find the value :** $\dfrac{\|\vec{A} \times \vec{B}\|^2 + |\vec{A} \,.\, \vec{B}|^2}{2\,\|\vec{A}\|^2 \,\|\vec{B}\|^2}$ « $\frac{1}{2}$ »

46 **Prove that :** $\|\vec{A} \times \vec{B}\| = \sqrt{(\vec{A} \,.\, \vec{A})\,(\vec{B} \,.\, \vec{B}) - (\vec{A} \,.\, \vec{B})^2}$

47 Use vectors to prove that the area of the coplanar quadrilateral ABCD is $\frac{1}{2}\,\|\overrightarrow{AC} \times \overrightarrow{BD}\|$

48 If $\vec{A}$ and $\vec{B}$ are two unit vectors in $\mathbb{R}^3$, at what condition does the result of the vector product $\vec{A} \times \vec{B}$ represent a unit vector in $\mathbb{R}^3$. Explain your answer.

49 If the sum of two unit vectors is a unit vector too , prove that the difference between them is a vector of norm $\sqrt{3}$

50 **Prove that for any three vectors A , B , C :**

(1) $(\vec{A} + \vec{B} - \vec{C}) \times (\vec{A} - \vec{B} + \vec{C}) = 2\,\vec{A} \times (\vec{C} - \vec{B})$

(2) $(\vec{A} - \vec{B} - \vec{C}) \times (\vec{A} + \vec{B} + \vec{C}) = 2\,\vec{A} \times (\vec{C} + \vec{B})$

Higher skills

51 If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three unit vectors , $\vec{C}$ is not in the same plane as $\vec{A}$ and $\vec{B}$ where the measure of the angle between the two vectors $\vec{A}$, $\vec{B}$ equals the measure of the angle between $(\vec{A} \times \vec{B})$, $\vec{C}$ equals θ

Prove that : $(\vec{A} \times \vec{B}) \,.\, \vec{C} = \frac{1}{2} \sin 2\,\theta$

52 If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three non parallel vectors in the same plane where $\vec{A} + \vec{B} + \vec{C} = \vec{O}$, **prove that :** $\vec{A} \times \vec{B} = \vec{B} \times \vec{C} = \vec{C} \times \vec{A}$ and explain that geometrically.

53 If $\vec{A} \neq \vec{B} \neq \vec{O}$, $\vec{A} \times [\vec{B} \times (\vec{A} \times \vec{B})] = \vec{O}$ **Prove that :** Either $\vec{A} \,//\, \vec{B}$ or $\vec{A} \perp \vec{B}$